

BUSINESS INSIGHT REPORT

Olist Brazilian E-Commerce Analytics Project

Analytical Lens: Customer Retention & Loyalty Analyst

Unique KPI: Repeat Customer Rate / Repeat Purchase Ratio

This report consolidates the completed SQL, Statistical Analysis, Power BI, AI-Augmented Analytics, and Phase 8 business interpretation into one executive-oriented document. It focuses on business findings, evidence, implications, and actions rather than repeating technical implementation steps.

1. Executive Summary

The analysis shows a marketplace with strong overall customer-experience and operational indicators, but a substantial customer-retention opportunity. The statistical analysis reports 93,099 one-time customers (96.88%) and 2,997 repeat customers (3.12%). This makes repeat purchasing the primary strategic opportunity selected for the project.

Average order revenue is 137.75 compared with a median of 86.90, indicating that higher-value orders influence the average. Average review score is 4.0864/5, while average delivery delay is -11.8769 days; in this project, negative delay represents delivery earlier than the estimated date. The Power BI analysis reports a 97.02% delivery rate.

The recommended direction is to improve post-purchase retention, diagnose barriers to repeat purchasing, and use product, seller, and regional contribution analysis to prioritize commercial resources.

2. Business Problem & Objectives

The project evaluates e-commerce performance across customers, revenue, products, sellers, customer experience, and operations. The objective is to convert transactional data into actionable business insights and recommendations.

Primary business objectives: understand customer purchase behaviour; evaluate revenue and payment trends; identify product and category contribution; assess seller performance; understand customer satisfaction; and evaluate delivery and order-lifecycle performance.

3. Dataset & Analytical Scope

Dataset	Rows
Customers	99,441
Orders	99,441
Order Items	112,650
Products	32,951
Payments	103,886
Reviews	99,222 in MySQL analysis
Sellers	3,095

The project scope uses customers, orders, order_items, products, payments, reviews, and sellers. Geolocation and category-translation datasets were excluded from the mandatory analytical scope.

4. Methodology

The completed workflow covered data cleaning and validation, Excel analysis, relational SQL analysis, descriptive and comparative statistics, Power BI data modeling and dashboarding, and AI-assisted interpretation. The final business interpretation prioritizes evidence from completed analyses and explicitly avoids unsupported causal claims.

5. Customer Intelligence

Finding: Customer activity is strongly concentrated among one-time purchasers. One-time customers account for 96.88% of the analyzed customer-type population, while repeat customers account for 3.12%.

Business implication: Increasing the number of customers alone may not maximize long-term value. A retention strategy should focus on converting first-time purchasers into repeat purchasers and measuring the result through repeat customer rate and subsequent order activity.

6. Revenue Intelligence

Average order revenue is 137.75 and median order revenue is 86.90. The gap indicates that higher-value orders influence the mean and that management should monitor both average and median performance.

Payment behaviour was analyzed by payment type and payment value. Monthly revenue and order trends were also incorporated into the Power BI dashboard to support time-based performance monitoring.

7. Product Intelligence

Product and category performance was evaluated through revenue, order volume, product count, and revenue share. This enables management to distinguish high-value categories from categories that primarily generate transaction volume.

8. Seller Intelligence

Seller performance was evaluated using revenue, order volume, seller state, and average order value. Top sellers by revenue and orders provide a practical view of marketplace contribution and can support targeted seller-management decisions.

9. Customer Experience

Average review score is 4.0864/5, indicating generally strong customer satisfaction. Review-score distribution and review performance by product category were analyzed, alongside review score versus delivery delay.

The analysis does not claim that delivery performance causes review outcomes because a formal causal analysis was not documented.

10. Operational Intelligence

Average delivery delay is -11.8769 days. Within the project definition, negative values represent delivery earlier than estimated. The Power BI dashboard reports a 97.02% delivery rate. Order-status distribution, delivered and cancelled orders, delayed orders, monthly order trends, and on-time delivery performance were also analyzed.

11. Analytical Differentiation

Primary lens: Customer Retention & Loyalty Analyst.

Unique KPI: Repeat Customer Rate / Repeat Purchase Ratio. It measures the percentage of customers who made more than one order during the analyzed period and directly aligns with the selected retention lens. The documented statistical customer-type analysis reports a 3.12% repeat-customer rate.

12. Business Hypotheses & Validation

Hypothesis	Evidence / Validation	Conclusion
H1: Repeat customers have a different order volume distribution performance. The repeat customers not contain the numerical data, statistical significance is not claimed.	Monthly order volume distribution performance were analyzed and visualized together.	Descriptive analysis, not causal/significant
H2: Delivery performance may be associated with customer satisfaction.	Delivery delay and customer satisfaction were analyzed and visualized together.	Descriptive analysis, not causal/significant
H3: Certain customer regions contribute disproportionately to state and customer distribution by state were analyzed and visualized.	Customer distribution by state were analyzed and visualized.	Descriptive analysis, not causal/significant; no formal concern

13. Key Analytical Insights

- 1. Retention is the clearest strategic opportunity:** the 3.12% repeat-customer rate indicates a large gap between acquisition and repeat activity.
- 2. Mean revenue can mask the underlying order distribution:** the 137.75 average versus 86.90 median shows that higher-value orders materially influence the mean.
- 3. Strong experience indicators do not automatically imply strong loyalty:** the 4.0864/5 average review score and generally strong delivery indicators coexist with a low repeat-customer share.
- Unexpected finding:** the documented descriptive means show one-time customers at 138.6204 versus 124.9113 for repeat customers. This does not prove repeat customers are less valuable overall because the documented significance result is unavailable.

14. Data-Driven Recommendations

Priority	Action	Expected Impact	Key Risk
1	Build targeted post-purchase retention initiatives for one-time customers	Increase repeat-customer rate, repeat orders, and potential customer lifetime value	poorly targeted campaigns
2	Diagnose post-purchase barriers using customer activity, review, delivery, and product evidence	Identify areas for improvement, enhance customer experience	Descriptive relationships may not establish causality with outcomes
3	Prioritize commercial resources using product, seller, and regional revenue/order analysis	Improve allocation of sales and marketing resources	Resource concentration can increase dependency on a limited set of products, sellers, or regions

15. If I Were the Business Head...

- Action 1 — Improve retention:** establish a measurable post-purchase retention program focused on converting one-time customers into repeat customers. Baseline: 3.12% repeat customer rate.
- Action 2 — Diagnose first-to-second-purchase barriers:** combine customer, review, delivery, and product signals to identify where the post-purchase journey can be improved. Track subsequent purchase activity.
- Action 3 — Allocate resources using contribution evidence:** use category, seller, and regional revenue/order analysis to prioritize commercial attention while monitoring concentration risk.

16. AI-Assisted Analysis & Validation

AI was used to generate alternative interpretations, challenge assumptions through counter-analysis prompts, refine recommendations, and support executive storytelling. AI outputs were treated as hypotheses and retained only when supported by SQL, statistical analysis, Power BI evidence, or explicit limitations.

A key validation control was avoiding unsupported statistical claims. In particular, no p-value or significance conclusion was invented for the Mann–Whitney U analysis because the documented report did not provide the numerical result.

17. Key Metrics Summary

Metric	Result
One-Time Customers	93,099 (96.88%)
Repeat Customers	2,997 (3.12%)
Average Order Revenue	137.75
Median Order Revenue	86.90
Average Review Score	4.0864 / 5
Average Delivery Delay	-11.8769 days
Delivery Rate	97.02%
Price–Freight Correlation	0.4142
Revenue Outlier Percentage	7.9575%

18. Limitations

- The documented Mann–Whitney U test does not include the numerical test statistic or p-value, so statistical significance is not claimed.
- Delivery and review analysis is descriptive; no causal relationship is established.
- Regional revenue analysis supports comparison, but a formal concentration statistic is not documented.
- The report focuses on the mandatory project datasets and completed analyses.

19. Final Conclusion

The project indicates that the strongest strategic opportunity is customer retention. The marketplace shows stronger operational and customer-satisfaction indicators than its repeat-purchase behaviour suggests. Management should therefore focus on converting first-time buyers into repeat customers, use customer-experience and operational signals to diagnose barriers, and prioritize commercial resources using evidence from product, seller, and regional contribution analysis. The Repeat Customer Rate provides a clear baseline for measuring progress.